

From Primitives to Berry Phase

A Walkthrough of the Event Density Derivation

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1 The Question

What this walkthrough derives

This walkthrough derives, from substrate primitives, three closely-related quantities of standard quantum mechanics:

1. **The Berry connection** $\mathcal{A}_n(\mathbf{R}) = i\langle n(\mathbf{R}) | \nabla_{\mathbf{R}} n(\mathbf{R}) \rangle$ on parameter space, as a substrate-level rule-type connection forced by parallel-transport compatibility with normalization and unitarity.
2. **The Berry phase** $\gamma_n[C] = \oint_C \mathcal{A}_n(\mathbf{R}) \cdot d\mathbf{R}$, as the geometric phase accumulated under adiabatic cyclic evolution around a closed loop C in parameter space.
3. **The Berry curvature** $\Omega_n(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathcal{A}_n(\mathbf{R})$, as the gauge-invariant curvature of the rule-type connection, satisfying $\gamma_n[C] = \iint_S \Omega_n \cdot d\mathbf{S}$ for any surface S with $\partial S = C$.

Each is derived, not posited. The walkthrough is fully self-contained: all required Hilbert-space, parallel-transport, and gauge-transformation algebra is derived inside the document.

Why standard quantum mechanics treats Berry phase as a formal construction

Berry's 1984 derivation observed that adiabatic cyclic evolution leaves an instantaneous eigenstate $|n(\mathbf{R})\rangle$ unchanged in identity but multiplied by a phase factor $e^{i\gamma_n}$ above and beyond the standard dynamical phase. The construction proceeds: write $|\psi(t)\rangle = e^{-i \int_0^t E_n(t') dt' / \hbar} e^{i\gamma_n(t)} |n(\mathbf{R}(t))\rangle$, substitute into the Schrödinger equation, project onto $\langle n|$, and read off $\dot{\gamma}_n = i\langle n | \dot{n} \rangle = i\langle n | \nabla_{\mathbf{R}} n \rangle \cdot \dot{\mathbf{R}}$.

This is mathematically correct. It is also mechanistically opaque. Standard QM does not say *what makes the parameter-dependent eigenstate $|n(\mathbf{R})\rangle$ a coherent object across $\mathbf{R}$ -values that are never simultaneously instantiated, why parallel transport produces a connection rather than a flat structure, what makes the geometric phase rate-independent, or what physical object the Berry curvature describes*. The construction is presented as a derived consequence of QM postulates, but the postulates themselves are unexplained at any deeper ontological level. The Berry connection and curvature are objects in parameter space, not physical space — their meaning at the level of underlying ontology is left unstated.

What Event Density claims

The Berry connection, Berry phase, and Berry curvature are FORCED by the substrate ontology of the Event Density (ED) framework when participation rules are parameter-dependent. The connection one-form is the substrate-level statement that adiabatic identity-preservation under parameter change requires a path-dependent compensation, the geometric-phase rate-independence is the substrate-level statement that the connection lives on parameter space rather than time, and the curvature is the substrate-level rule-type curvature in parameter space

— directly analogous to the rule-type curvature of a gauge field in physical space, but lifted to the space of rule parameters.

The substrate-level mechanism, in one sentence: **the Berry phase is the holonomy of the rule-type connection on the bundle of parameter-dependent participation rules.**

The chain in summary

The derivation chain runs:

substrate primitives with explicit parameter dependence (§2) → parameter-dependent participation rule + instantaneous eigenchannels (§3) → parallel-transport condition forces the Berry connection (§4) → closed adiabatic loop separates dynamical and geometric phase (§5) → Berry curvature as gauge-invariant rule-type curvature, with Stokes-theorem relation (§6) → substrate-level reading + explicit comparison with the closed-loop integral structure familiar from gauge fields in physical space (§7) → exact claims established (§8).

The substrate primitives are five: participation rule with parameter dependence, identity alignment with pre-individuation amplitudes, channels and adiabatic evolution as slow parameter change, global ED-geometry in parameter space, and closed loops in parameter space as closed families of participation rules.

2 The Primitives

2.1 P-BP-1. Parameter-dependent participation rule

A *participation rule* $r(\mathbf{R})$ is the substrate-level identity-encoding of a chain, *with explicit functional dependence on a parameter vector* $\mathbf{R} \in \mathcal{M}$, where $\mathcal{M}$ is a smooth manifold (the parameter manifold). At each fixed $\mathbf{R}$, the rule is a single-typed object: a chain at parameter $\mathbf{R}$ commits to exactly one rule drawn from a discrete alignment set $\mathcal{R}(\mathbf{R}) = \{r_0(\mathbf{R}), r_1(\mathbf{R}), \dots\}$.

Smooth dependence. As $\mathbf{R}$ varies smoothly, the alignment set varies smoothly: each $r_n(\mathbf{R})$ is a smooth function of $\mathbf{R}$ in the substrate-level sense that the inner-product structure $\langle r_i(\mathbf{R}) | r_j(\mathbf{R}) \rangle$ is smoothly differentiable in $\mathbf{R}$.

Algebraic structure. The alignment set carries a substrate-level inner-product structure with $\langle r_i(\mathbf{R}) | r_i(\mathbf{R}) \rangle = 1$ and $\langle r_i(\mathbf{R}) | r_j(\mathbf{R}) \rangle = 0$ for $i \neq j$ (orthonormal alignment set at each $\mathbf{R}$). This is the parameter-space generalization of the inner-product structure on a fixed alignment set.

Pre-individuation amplitudes at fixed $\mathbf{R}$. Before commitment at parameter $\mathbf{R}$, the chain admits multiple consistent rule continuations weighted by complex amplitudes $\alpha_n(\mathbf{R}) \in \mathbb{C}$:

$$|\psi(\mathbf{R})\rangle = \sum_n \alpha_n(\mathbf{R}) |r_n(\mathbf{R})\rangle, \quad \sum_n |\alpha_n(\mathbf{R})|^2 = 1.$$

2.2 P-BP-2. Identity alignment under parameter change

The *identity alignment* of a chain at parameter $\mathbf{R}$ is the chain's commitment to one specific rule $r_n(\mathbf{R})$ within the alignment set. As $\mathbf{R}$ changes, the rule $r_n(\mathbf{R})$ deforms continuously along the parameter trajectory.

Identity-preserving evolution. A chain whose alignment is $r_n(\mathbf{R})$ at $\mathbf{R} = \mathbf{R}_0$ may continue to be aligned with the *same labeled rule* $r_n(\mathbf{R})$ as $\mathbf{R}$ varies smoothly to $\mathbf{R}_1$, provided the evolution is slow enough not to force a transition to a different rule $r_m(\mathbf{R})$ with $m \neq n$. This is the substrate-level statement of adiabatic identity preservation.

Pre-individuation amplitudes carry over. A pre-individuation state $|\psi(\mathbf{R})\rangle = \sum_n \alpha_n(\mathbf{R}) |r_n(\mathbf{R})\rangle$ that is concentrated entirely on the n -th rule (i.e., $\alpha_n(\mathbf{R}_0) = 1$, others zero) remains so under

identity-preserving evolution: $\alpha_n(\mathbf{R}(t)) = \alpha_n(\mathbf{R}_0)$ in magnitude, modulo a phase factor whose structure is the subject of §4.

2.3 P-BP-3. Channel and adiabatic evolution

A *channel* is a participation pathway through ED-gradients carrying a single alignment thread. At each substrate-tick the channel commits to one rule from the alignment set $\mathcal{R}(\mathbf{R})$ at the current parameter.

Adiabatic evolution. A *slow* parameter change is a trajectory $\mathbf{R}(t)$ along which the parameter changes on timescales T much longer than the substrate-level commitment timescale set by the gap between $r_n(\mathbf{R})$ and the nearest other rule $r_m(\mathbf{R})$:

$$T \gg \hbar/\Delta_{nm}(\mathbf{R}), \quad \Delta_{nm}(\mathbf{R}) = E_n(\mathbf{R}) - E_m(\mathbf{R}) \quad (m \neq n)$$

is the energy gap (in standard QM coarse-grained units; defined in §3). Under adiabatic evolution, the channel's identity stays committed to the n -th rule throughout the parameter trajectory.

Substrate-level statement. Adiabatic evolution is the regime in which the parameter manifold's geometric structure becomes dominant: the channel cannot commit fast enough to follow rapid local fluctuations, and its commitment is forced to track the smooth deformation of the rule itself.

2.4 P-BP-4. Global ED-geometry in parameter space

The *global ED-geometry in parameter space* is the structure imposed on the parameter manifold $\mathcal{M}$ by the family of alignment sets $\{\mathcal{R}(\mathbf{R})\}_{\mathbf{R} \in \mathcal{M}}$. Each point $\mathbf{R}$ carries a Hilbert-space-like fibre (the alignment set + amplitudes), and the smooth dependence of the alignment set on $\mathbf{R}$ defines a fibre bundle over $\mathcal{M}$.

Fibre bundle structure. The total space is the disjoint union $E = \bigsqcup_{\mathbf{R} \in \mathcal{M}} \mathcal{H}(\mathbf{R})$, where $\mathcal{H}(\mathbf{R})$ is the Hilbert space spanned by $\{|r_n(\mathbf{R})\rangle\}$. The projection $\pi : E \rightarrow \mathcal{M}$ sends each $|\psi(\mathbf{R})\rangle$ to its base point $\mathbf{R}$. The fibre at $\mathbf{R}$ is $\mathcal{H}(\mathbf{R})$.

The bundle is generally non-trivial. The choice of basis $\{|r_n(\mathbf{R})\rangle\}$ at each $\mathbf{R}$ may not extend to a globally smooth choice across all of $\mathcal{M}$. Different choices differ by $\mathbf{R}$ -dependent phase factors $e^{i\chi_n(\mathbf{R})}$. This non-triviality is what makes the Berry phase non-zero in general.

2.5 P-BP-5. Closed loop in parameter space

A *closed loop* C in parameter space is a smooth path $\mathbf{R} : [0, T] \rightarrow \mathcal{M}$ with $\mathbf{R}(T) = \mathbf{R}(0)$. The parameter manifold returns to its starting point; the alignment set $\mathcal{R}(\mathbf{R}(T)) = \mathcal{R}(\mathbf{R}(0))$ is identical to its initial form.

Substrate-level statement. A closed loop in parameter space is a closed family of participation rules: a one-parameter family of rules that returns to its starting member after one cycle. The chain's identity, if preserved adiabatically, returns to the same committed rule. The non-trivial question — addressed in §5 — is whether the chain's pre-individuation amplitude returns to its initial value or accumulates a phase.

3 From Parameter-Dependent Rules to Instantaneous Eigenchannels

3.1 Hamiltonian as coarse-grained parameter-dependent rule

The parameter-dependent participation rule, coarse-grained via the substrate-to-Hilbert-space bridge, produces a parameter-dependent Hamiltonian operator $H(\mathbf{R})$ on the fibre Hilbert space

$\mathcal{H}(\mathbf{R})$. The eigenvalue equation

$$H(\mathbf{R})|n(\mathbf{R})\rangle = E_n(\mathbf{R})|n(\mathbf{R})\rangle$$

defines instantaneous eigenchannels $|n(\mathbf{R})\rangle$ with instantaneous eigen-energies $E_n(\mathbf{R})$. The eigenchannel set $\{|n(\mathbf{R})\rangle\}$ is an orthonormal basis for $\mathcal{H}(\mathbf{R})$:

$$\langle m(\mathbf{R})|n(\mathbf{R})\rangle = \delta_{mn}.$$

The eigenchannels are the coarse-grained images of the parameter-dependent participation rules: $|n(\mathbf{R})\rangle$ corresponds to the substrate-level rule $r_n(\mathbf{R})$.

3.2 Smooth dependence

The Hamiltonian $H(\mathbf{R})$ is a smooth function of $\mathbf{R}$ — its matrix elements are smooth in any fixed basis — and the eigen-energies $E_n(\mathbf{R})$ are smooth functions on the parameter manifold (away from level crossings). The eigenchannels $|n(\mathbf{R})\rangle$ are smooth functions of $\mathbf{R}$ *up to a phase*: the eigenvalue equation determines $|n(\mathbf{R})\rangle$ only up to multiplication by an arbitrary phase $e^{i\chi_n(\mathbf{R})}$. This phase ambiguity is the source of the gauge structure on parameter space (§6).

3.3 Adiabatic evolution at the Hilbert-space level

The full Schrödinger equation under time-varying parameter $\mathbf{R}(t)$:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H(\mathbf{R}(t))|\psi(t)\rangle.$$

Expand $|\psi(t)\rangle$ in the instantaneous eigenchannel basis:

$$|\psi(t)\rangle = \sum_n c_n(t) e^{-i\varphi_n(t)} |n(\mathbf{R}(t))\rangle,$$

where $\varphi_n(t) = \frac{1}{\hbar} \int_0^t E_n(\mathbf{R}(t')) dt'$ is the dynamical phase. Substituting into Schrödinger and projecting onto $\langle m(\mathbf{R}(t))|$:

$$\dot{c}_m(t) = - \sum_n c_n(t) e^{i(\varphi_m - \varphi_n)} \langle m(\mathbf{R}(t)) | \frac{d}{dt} | n(\mathbf{R}(t)) \rangle.$$

The matrix element $\langle m | \frac{d}{dt} | n \rangle = \langle m | \nabla_{\mathbf{R}} n \rangle \cdot \dot{\mathbf{R}}$. For $m \neq n$, the standard adiabatic argument bounds this term by

$$|\langle m | \nabla_{\mathbf{R}} n \rangle \cdot \dot{\mathbf{R}}| \lesssim |\dot{\mathbf{R}}| / \Delta_{mn}(\mathbf{R}),$$

(from the off-diagonal Hellmann–Feynman identity, derived in §4.2 below). The phase factor $e^{i(\varphi_m - \varphi_n)}$ oscillates rapidly under the adiabatic condition $|\dot{\mathbf{R}}| / \Delta_{mn} \ll 1$, and the off-diagonal contribution to $\dot{c}_m$ averages to zero on adiabatic timescales.

Conclusion of the adiabatic argument. Under adiabatic evolution, $|c_n(t)|$ is preserved: a chain initially in the n -th instantaneous eigenchannel remains in the n -th instantaneous eigenchannel throughout the evolution. The remaining content of $c_n(t)$ is its phase, addressed in §4.

4 Parallel Transport and the Berry Connection

4.1 The diagonal phase equation

For a chain initially in $|n(\mathbf{R}(0))\rangle$, adiabatic evolution maintains $|c_n(t)| = 1$ and $|c_m(t)| = 0$ for $m \neq n$. The state at time t is therefore

$$|\psi(t)\rangle = c_n(t) e^{-i\varphi_n(t)} |n(\mathbf{R}(t))\rangle$$

with $|c_n(t)| = 1$, so $c_n(t) = e^{i\gamma_n(t)}$ for some real $\gamma_n(t)$. The diagonal Schrödinger projection gives:

$$\dot{c}_n(t) = -c_n(t) \langle n(\mathbf{R}(t)) | \frac{d}{dt} | n(\mathbf{R}(t)) \rangle,$$

which yields, for $c_n(t) = e^{i\gamma_n(t)}$:

$$i\dot{\gamma}_n(t) = -\langle n(\mathbf{R}(t)) | \frac{d}{dt} | n(\mathbf{R}(t)) \rangle$$

equivalently

$$\dot{\gamma}_n(t) = i \langle n(\mathbf{R}(t)) | \frac{d}{dt} | n(\mathbf{R}(t)) \rangle = i \langle n(\mathbf{R}(t)) | \nabla_{\mathbf{R}} n(\mathbf{R}(t)) \rangle \cdot \dot{\mathbf{R}}(t).$$

4.2 The off-diagonal Hellmann–Feynman identity

We pause to derive the off-diagonal identity used in §3.3. Differentiating the eigenvalue equation $H|n\rangle = E_n|n\rangle$ with respect to $\mathbf{R}$:

$$(\nabla_{\mathbf{R}} H)|n\rangle + H|\nabla_{\mathbf{R}} n\rangle = (\nabla_{\mathbf{R}} E_n)|n\rangle + E_n|\nabla_{\mathbf{R}} n\rangle.$$

Project onto $\langle m|$ for $m \neq n$:

$$\langle m|\nabla_{\mathbf{R}} H|n\rangle + \langle m|H|\nabla_{\mathbf{R}} n\rangle = (\nabla_{\mathbf{R}} E_n)\langle m|n\rangle + E_n\langle m|\nabla_{\mathbf{R}} n\rangle.$$

Using $\langle m|H = E_m\langle m|$ and $\langle m|n\rangle = 0$:

$$\langle m|\nabla_{\mathbf{R}} H|n\rangle + E_m\langle m|\nabla_{\mathbf{R}} n\rangle = E_n\langle m|\nabla_{\mathbf{R}} n\rangle,$$

which gives

$$\langle m|\nabla_{\mathbf{R}} n\rangle = \frac{\langle m|\nabla_{\mathbf{R}} H|n\rangle}{E_n - E_m} \quad \text{for } m \neq n.$$

This is the off-diagonal Hellmann–Feynman identity. It bounds $\langle m|\nabla_{\mathbf{R}} n\rangle$ by $\|\nabla_{\mathbf{R}} H\|/\Delta_{mn}$, confirming the adiabatic suppression argument.

4.3 Normalization constraint forces $\langle n|\nabla_{\mathbf{R}} n\rangle$ to be imaginary

Differentiate $\langle n(\mathbf{R})|n(\mathbf{R})\rangle = 1$ with respect to $\mathbf{R}$:

$$\langle \nabla_{\mathbf{R}} n|n\rangle + \langle n|\nabla_{\mathbf{R}} n\rangle = 0,$$

which gives

$$\langle n|\nabla_{\mathbf{R}} n\rangle = -\langle \nabla_{\mathbf{R}} n|n\rangle = -\langle n|\nabla_{\mathbf{R}} n\rangle^*,$$

so $\langle n|\nabla_{\mathbf{R}} n\rangle$ is purely imaginary. Therefore $i\langle n|\nabla_{\mathbf{R}} n\rangle$ is purely real.

4.4 The Berry connection

Define the **Berry connection** $\mathcal{A}_n(\mathbf{R})$ on the parameter manifold:

$$\mathcal{A}_n(\mathbf{R}) \equiv i\langle n(\mathbf{R})|\nabla_{\mathbf{R}} n(\mathbf{R})\rangle.$$

By the normalization argument in §4.3, $\mathcal{A}_n(\mathbf{R})$ is a real-valued vector field on $\mathcal{M}$. By the diagonal phase equation in §4.1, it is the rate at which the geometric phase accumulates per unit parameter-space displacement:

$$\dot{\gamma}_n(t) = \mathcal{A}_n(\mathbf{R}(t)) \cdot \dot{\mathbf{R}}(t).$$

Integrating from 0 to T :

$$\gamma_n(T) - \gamma_n(0) = \int_0^T \mathcal{A}_n(\mathbf{R}(t)) \cdot \dot{\mathbf{R}}(t) dt = \int_C \mathcal{A}_n(\mathbf{R}) \cdot d\mathbf{R},$$

where C is the parameter-space path traced by $\mathbf{R}(t)$.

This is the Berry connection, derived not posited. It is forced by three substrate-level requirements:

- **Smooth parameter dependence** (P-BP-1) gives well-defined $\nabla_{\mathbf{R}}|n(\mathbf{R})\rangle$.
- **Normalization** of the eigenchannel forces $\langle n|\nabla_{\mathbf{R}}n\rangle$ to be purely imaginary, so $\mathcal{A}_n$ is real.
- **Adiabatic identity preservation** (P-BP-2 + P-BP-3) restricts the evolution to the diagonal sector, isolating the diagonal phase rate as the only non-trivial content.

4.5 Parallel-transport reading

The condition for *parallel transport* of the chain's identity along a path $\mathbf{R}(t)$ is that no “extra” phase be acquired beyond what is forced by the rule-type structure. Equivalently, the chain's pre-individuation state in the n -th eigenchannel is

$$|\psi_{\text{PT}}(t)\rangle \equiv e^{i\gamma_n(t)}|n(\mathbf{R}(t))\rangle,$$

with $\gamma_n(t)$ determined by the requirement that $\langle \psi_{\text{PT}}(t)|\nabla_{\mathbf{R}}|\psi_{\text{PT}}(t)\rangle$ be zero modulo the Berry-connection compensation:

$$\langle \psi_{\text{PT}}|\frac{d}{dt}|\psi_{\text{PT}}\rangle = i\dot{\gamma}_n + \langle n|\frac{d}{dt}|n\rangle = i\dot{\gamma}_n - i\mathcal{A}_n \cdot \dot{\mathbf{R}} = 0.$$

So $\gamma_n(t) = \int_0^t \mathcal{A}_n(\mathbf{R}(t')) \cdot \dot{\mathbf{R}}(t') dt'$ is exactly the rate that compensates for the parameter-induced rotation of the eigenchannel within the fibre Hilbert space. This is the substrate-level statement that the Berry connection is the connection on the bundle defined in P-BP-4 — the structure that defines parallel transport of the rule-type identity.

5 Closed Loops and the Berry Phase

5.1 Setup

Consider a closed loop C in parameter space: $\mathbf{R}(t)$ for $t \in [0, T]$ with $\mathbf{R}(T) = \mathbf{R}(0) \equiv \mathbf{R}_0$. The chain begins in the instantaneous n -th eigenchannel: $|\psi(0)\rangle = |n(\mathbf{R}_0)\rangle$.

5.2 State at time T

By the adiabatic argument (§3.3) the chain remains in the instantaneous n -th eigenchannel throughout. By §4.4 it accumulates a phase consisting of two pieces:

$$|\psi(T)\rangle = e^{-i\varphi_n(T)} e^{i\gamma_n(T)} |n(\mathbf{R}(T))\rangle = e^{-i\varphi_n(T)} e^{i\gamma_n(T)} |n(\mathbf{R}_0)\rangle.$$

The eigenchannel at the end of the loop is the same labeled rule as at the start (P-BP-5: closed loop in parameter space returns the alignment set to itself), and the chain's commitment is to the same rule throughout (adiabaticity). The non-trivial content lives entirely in the phase.

5.3 Dynamical phase

The dynamical phase is

$$\varphi_n(T) = \frac{1}{\hbar} \int_0^T E_n(\mathbf{R}(t)) dt.$$

This phase depends on the *speed* of traversal: a slower traversal accumulates a larger dynamical phase because it spends more time at each instantaneous eigen-energy. The dynamical phase is *not* a geometric quantity — it is an integral of E_n over time, and reparameterizing t changes its value.

5.4 Geometric (Berry) phase

The geometric phase is

$$\gamma_n(T) = \int_0^T \mathcal{A}_n(\mathbf{R}(t)) \cdot \dot{\mathbf{R}}(t) dt = \oint_C \mathcal{A}_n(\mathbf{R}) \cdot d\mathbf{R}.$$

The right-hand side is a *line integral over the closed loop in parameter space*. This integral does *not* depend on the speed of traversal: $\dot{\mathbf{R}} dt = d\mathbf{R}$ is a parameter-space differential, and the line integral $\oint_C \mathcal{A}_n(\mathbf{R}) \cdot d\mathbf{R}$ is determined entirely by the geometric path C in parameter space — not by how fast the chain traverses it. **The Berry phase is rate-independent because it is a geometric invariant of the path in parameter space.**

This is the central derived result:

$$\gamma_n[C] = \oint_C \mathcal{A}_n(\mathbf{R}) \cdot d\mathbf{R}, \quad \mathcal{A}_n(\mathbf{R}) = i\langle n(\mathbf{R}) | \nabla_{\mathbf{R}} n(\mathbf{R}) \rangle.$$

5.5 Substrate-level reading

- The chain's commitment to the n -th rule is preserved throughout the closed loop (adiabaticity + identity-preserving evolution).
- The chain's pre-individuation amplitude returns to the initial labeled rule but with an accumulated phase.
- The phase splits into a dynamical part (depending on the rate of traversal) and a geometric part (depending only on the path in parameter space).
- The geometric part is the integrated Berry connection — the holonomy of the rule-type connection on the parameter-space bundle.

5.6 The Berry phase as global participation twist

The substrate-level meaning: the family of participation rules indexed by the closed loop C is a *closed family* whose first member equals its last. The chain that traverses the family has its identity-commitment preserved at every step, but the family itself has a global twist. The Berry phase measures this twist as the phase mismatch between the chain's pre-individuation amplitude and the labeled-rule basis at the loop's closure.

If the bundle in P-BP-4 is trivial — the basis $\{|n(\mathbf{R})\rangle\}$ extends to a globally smooth choice across the entire loop — then $\mathcal{A}_n$ is a pure gradient and $\oint_C \mathcal{A}_n \cdot d\mathbf{R} = 0$ (modulo 2π). If the bundle is non-trivial, the gradient cannot be globally defined, and the loop integral is generically non-zero.

6 Berry Curvature and Stokes' Theorem

6.1 Gauge transformations of the Berry connection

The eigenchannel $|n(\mathbf{R})\rangle$ is determined by the eigenvalue equation only up to a phase: $|n(\mathbf{R})\rangle$ and

$$|\tilde{n}(\mathbf{R})\rangle \equiv e^{i\chi(\mathbf{R})} |n(\mathbf{R})\rangle$$

both satisfy $H(\mathbf{R})|\cdot\rangle = E_n(\mathbf{R})|\cdot\rangle$, for any smooth real-valued function $\chi(\mathbf{R})$ on $\mathcal{M}$. Compute the Berry connection in the new basis:

$$\begin{aligned}\tilde{\mathcal{A}}_n(\mathbf{R}) &= i\langle\tilde{n}(\mathbf{R})|\nabla_{\mathbf{R}}\tilde{n}(\mathbf{R})\rangle \\ &= i\langle n(\mathbf{R})|e^{-i\chi(\mathbf{R})}\nabla_{\mathbf{R}}[e^{i\chi(\mathbf{R})}|n(\mathbf{R})\rangle] \\ &= i\langle n(\mathbf{R})|e^{-i\chi(\mathbf{R})}[i(\nabla_{\mathbf{R}}\chi)e^{i\chi(\mathbf{R})}|n(\mathbf{R})\rangle + e^{i\chi(\mathbf{R})}|\nabla_{\mathbf{R}}n(\mathbf{R})\rangle] \\ &= -\langle n(\mathbf{R})|n(\mathbf{R})\rangle\nabla_{\mathbf{R}}\chi + i\langle n(\mathbf{R})|\nabla_{\mathbf{R}}n(\mathbf{R})\rangle \\ &= \mathcal{A}_n(\mathbf{R}) - \nabla_{\mathbf{R}}\chi(\mathbf{R})\end{aligned}$$

(using the standard sign convention; some texts use the opposite sign for χ). The Berry connection transforms as a $U(1)$ gauge potential under the eigenchannel-phase rotation:

$$\mathcal{A}_n \rightarrow \tilde{\mathcal{A}}_n = \mathcal{A}_n - \nabla_{\mathbf{R}}\chi.$$

(Note the sign: this is the standard transformation law for a connection one-form under a $\chi \rightarrow e^{i\chi}$ gauge transformation; the sign convention varies in the literature. Here we use the convention where the Berry-phase loop integral is $+\oint \mathcal{A}_n \cdot d\mathbf{R}$.)

6.2 Gauge dependence of the Berry phase

Under the same gauge transformation, the Berry phase around a loop C transforms as

$$\tilde{\gamma}_n[C] = \oint_C \tilde{\mathcal{A}}_n \cdot d\mathbf{R} = \oint_C \mathcal{A}_n \cdot d\mathbf{R} - \oint_C \nabla_{\mathbf{R}}\chi \cdot d\mathbf{R} = \gamma_n[C] - [\chi(\mathbf{R}(T)) - \chi(\mathbf{R}(0))].$$

For a closed loop, $\mathbf{R}(T) = \mathbf{R}(0)$, so $\chi(\mathbf{R}(T)) - \chi(\mathbf{R}(0)) = 0$ if χ is a single-valued smooth function on $\mathcal{M}$. **In that case, the Berry phase is gauge-invariant:** $\tilde{\gamma}_n[C] = \gamma_n[C]$.

If χ is multi-valued (e.g., χ has $2\pi n$ winding around C), the Berry phase shifts by $2\pi n$ — but $e^{i\gamma_n[C]}$, the physically observable phase factor, is still gauge-invariant. The Berry phase is gauge-invariant modulo 2π .

6.3 The Berry curvature

Define the **Berry curvature** as the exterior derivative (curl) of the Berry connection:

$$\Omega_n(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathcal{A}_n(\mathbf{R}) \quad (\text{in three-dimensional parameter space})$$

or, more generally, as the two-form

$$\Omega_n = d\mathcal{A}_n = \frac{1}{2}(\partial_i \mathcal{A}_{n,j} - \partial_j \mathcal{A}_{n,i}) dR^i \wedge dR^j,$$

where $\mathcal{A}_n = \mathcal{A}_{n,i} dR^i$.

Gauge invariance of Ω_n . Under $\mathcal{A}_n \rightarrow \mathcal{A}_n - \nabla\chi$:

$$\tilde{\Omega}_n = d(\mathcal{A}_n - d\chi) = d\mathcal{A}_n - d^2\chi = d\mathcal{A}_n = \Omega_n,$$

since $d^2 = 0$ for any smooth χ . The Berry curvature is gauge-invariant *unconditionally*, not merely modulo 2π .

6.4 Stokes' theorem and the integrated curvature

For any surface $S \subset \mathcal{M}$ with boundary $\partial S = C$, Stokes' theorem on the parameter manifold gives:

$$\oint_C \mathcal{A}_n \cdot d\mathbf{R} = \iint_S (\nabla_{\mathbf{R}} \times \mathcal{A}_n) \cdot d\mathbf{S} = \iint_S \Omega_n \cdot d\mathbf{S},$$

provided $\mathcal{A}_n$ is smooth on S (the bundle is trivializable over S). Therefore:

$$\gamma_n[C] = \iint_S \Omega_n \cdot d\mathbf{S}.$$

The Berry phase around C is the integral of the Berry curvature over any surface S bounded by C . **This is the substrate-level Stokes-theorem relation: the holonomy of the rule-type connection equals the integrated curvature it bounds.**

6.5 Substrate-level reading of the curvature

- $\mathcal{A}_n(\mathbf{R})$ is the substrate-level *connection one-form* on the parameter manifold's bundle of participation rules.
- $\Omega_n(\mathbf{R})$ is the substrate-level *curvature two-form* of that connection.
- The Berry phase $\gamma_n[C]$ is the holonomy of the connection — the substrate-level statement of how much the rule-type identity has twisted around the closed loop.
- Gauge-invariance of Ω_n is the substrate-level statement that the curvature is a property of the family of participation rules itself, not of any choice of basis representation.

6.6 Surface independence

For two surfaces S_1 and S_2 both bounded by C , with $S_1 \cup (-S_2)$ enclosing a closed region V :

$$\iint_{S_1} \Omega_n \cdot d\mathbf{S} - \iint_{S_2} \Omega_n \cdot d\mathbf{S} = \oint_{\partial V} \mathcal{A}_n \cdot d\mathbf{R} - \oint_{\partial V} \mathcal{A}_n \cdot d\mathbf{R} = 0,$$

modulo possible gauge-related contributions if the bundle is not trivializable over V . In simply-connected regions of the parameter manifold, the integrated Berry curvature is surface-independent.

When the parameter manifold contains topologically non-trivial structures — degeneracy points where the bundle becomes singular — the Berry curvature integrated over a closed surface enclosing such a point yields a non-zero integer multiple of 2π (the Chern number of the bundle restricted to the surface). This integer-quantization is FORM-FORCED by the requirement that $\oint_C \mathcal{A}_n \cdot d\mathbf{R}$ be well-defined modulo 2π for any closed surface considered as the boundary of two complementary regions.

7 Substrate-Level Reading

The objects derived in §§4–6 admit the following substrate-level dictionary:

Standard QM object	Substrate-level meaning
Parameter-dependent eigenstate $ n(\mathbf{R})\rangle$	Coarse-grained image of parameter-dependent rule $r_n(\mathbf{R})$ (P-BP-1)
Berry connection $\mathcal{A}_n(\mathbf{R})$	Rule-type connection on the parameter-manifold bundle (P-BP-4)
Berry curvature $\Omega_n(\mathbf{R})$	Curvature of the rule-type connection in parameter space
Berry phase $\gamma_n[C]$	Holonomy of the rule-type connection: global participation twist accumulated along a closed family of rules
Gauge transformation $ n\rangle \rightarrow e^{i\chi} n\rangle$	Local re-phasing of the labeled-rule basis at each parameter point
Adiabatic theorem	Substrate-level identity-preservation under slow parameter change (P-BP-2 + P-BP-3)

7.1 Comparison with closed-loop integrals in physical space

The structure derived here — a connection one-form $\mathcal{A}$ on a manifold, a curvature two-form $\Omega = d\mathcal{A}$, a closed-loop integral $\oint_C \mathcal{A} \cdot dx$ producing a holonomy phase, gauge-invariance of the phase modulo 2π , and Stokes-theorem relation $\oint_C \mathcal{A} = \iint_S \Omega$ — is the same structure that appears for gauge fields propagating in physical space. The mathematical formalism is identical; the physical setting differs:

- **Physical-space gauge field:** $\mathcal{A}_\mu(x)$ defined on physical spacetime; closed-loop integral $\oint_C \mathcal{A}_\mu dx^\mu$ is the gauge phase accumulated by a charged chain traversing a physical loop C that may enclose flux-carrying regions.
- **Berry connection (this walkthrough):** $\mathcal{A}_n(\mathbf{R})$ defined on parameter space; closed-loop integral $\oint_C \mathcal{A}_n \cdot d\mathbf{R}$ is the geometric phase accumulated by an adiabatically-evolved chain traversing a closed loop in parameter space.

The substrate-level statement of the analogy: **the rule-type connection structure is the same in both cases — the difference is only the manifold on which the connection lives.** In one case it is physical spacetime; in the other it is the parameter manifold of rule-defining quantities. The substrate primitive is the rule-type connection; physical space and parameter space are two manifolds on which it can be instantiated.

7.2 The bundle is generally non-trivial

The non-triviality of the bundle in P-BP-4 — the fact that the basis $\{|n(\mathbf{R})\rangle\}$ cannot generally be extended smoothly across all of $\mathcal{M}$ — is the substrate-level origin of non-zero Berry phase. If the family of participation rules deforms smoothly around a closed loop without any topological obstruction, the Berry phase is zero (modulo 2π). If the family encircles a degeneracy point (where two eigen-rules become indistinguishable and the bundle structure becomes singular), the Berry phase is non-zero.

This is the substrate-level mechanism behind several standard observations: monopole-like Berry curvature near level crossings, π Berry phase for spin-1/2 rotations through 2π , and integer-valued Chern numbers for closed-surface-integrated Berry curvature.

8 What's Forced, What's Inherited, What's Open

What is FORCED by the substrate ontology

- Berry connection $\mathcal{A}_n(\mathbf{R}) = i\langle n(\mathbf{R}) | \nabla_{\mathbf{R}} n(\mathbf{R}) \rangle$ is FORCED by adiabatic evolution + normalization + smooth parameter dependence (§4).

- Berry phase $\gamma_n[C] = \oint_C \mathcal{A}_n \cdot d\mathbf{R}$ is FORCED as the geometric component of the total accumulated phase under adiabatic cyclic evolution (§5).
- Berry phase rate-independence is FORCED by the line-integral-over-parameter-space structure (§5.4).
- Berry curvature $\Omega_n = d\mathcal{A}_n$ and its gauge-invariance are FORCED by the differential-form structure of the connection (§6.3).
- Stokes-theorem relation $\gamma_n[C] = \iint_S \Omega_n \cdot d\mathbf{S}$ is FORCED for any surface S on which the bundle is trivializable (§6.4).
- The structural identity between the Berry connection and gauge connections in physical space is FORCED by the rule-type connection being a single substrate primitive instantiated on different manifolds (§7.1).

What is FORM-FORCED-INHERITED (and re-derived inside this document)

These are standard pieces of QM machinery that the substrate ontology reproduces and that this walkthrough re-derives:

- Hilbert-space inner product and orthonormality of eigenstates (§3.1).
- Schrödinger equation as the coarse-grained substrate-level evolution (§3.3).
- Off-diagonal Hellmann–Feynman identity $\langle m | \nabla_{\mathbf{R}} n \rangle = \langle m | \nabla_{\mathbf{R}} H | n \rangle / (E_n - E_m)$ (§4.2).
- Adiabatic theorem (preservation of $|c_n|$ under slow parameter evolution) (§3.3).
- Normalization derivative identity $\langle n | \nabla_{\mathbf{R}} n \rangle = -\langle n | \nabla_{\mathbf{R}} n \rangle^*$ (§4.3).
- Stokes’ theorem on smooth manifolds (§6.4).
- Differential-form algebra ($d^2 = 0$) for gauge invariance of curvature (§6.3).

What remains OPEN

- **Non-Abelian Berry connection** for degenerate eigenchannel subspaces. When two or more eigen-rules share an eigen-energy, the connection generalizes from a $U(1)$ one-form to a $U(N)$ matrix-valued one-form. Standard derivation (Wilczek–Zee) is FORM-FORCED-INHERITED; substrate-level reading parallels the non-Abelian extension in the gauge-fields walkthrough. Not derived here; OPEN as a candidate Wu–Yang / non-Abelian Berry walkthrough.
- **Sub-adiabatic corrections.** The leading-order adiabatic argument neglects $O(|\dot{\mathbf{R}}|/\Delta_{nm})$ corrections from off-diagonal mixing. FORM-FORCED expected; explicit substrate-level coefficient OPEN.
- **Geometric phase under non-cyclic evolution** (Aharonov–Anandan generalization). Berry’s derivation requires the loop to close; the Aharonov–Anandan generalization removes the cyclic requirement and produces a phase associated with any path in projective Hilbert space. Substrate-level reading parallel to the cyclic case; OPEN.
- **Berry phase for open systems.** Mixed states and Lindblad evolution generalize Berry phase to the geometric phase of Sjöqvist–Pati and Tong et al. FORM-FORCED-INHERITED at the standard-QM level; substrate-level reading via the Lindblad walkthrough is OPEN.
- **Quantization of integrated Berry curvature.** §6.6 stated the integer-quantization on closed surfaces enclosing degeneracy points without proof. Standard proof uses the Chern theorem; substrate-level reading OPEN as a candidate Chern-quantization walkthrough.

9 What This Argument Establishes

This walkthrough establishes the following exact claims:

Claim 1. The Berry connection $\mathcal{A}_n(\mathbf{R}) = i\langle n(\mathbf{R}) | \nabla_{\mathbf{R}} n(\mathbf{R}) \rangle$ is FORCED on parameter space by the combination of smooth parameter dependence (P-BP-1), adiabatic identity preservation (P-BP-2 + P-BP-3), and normalization of the eigenchannel. It is real-valued and transforms as a $U(1)$ connection under eigenchannel-phase gauge transformations.

Claim 2. The Berry phase $\gamma_n[C] = \oint_C \mathcal{A}_n \cdot d\mathbf{R}$ around a closed loop C in parameter space is FORCED as the geometric component of the total accumulated phase under adiabatic cyclic evolution. It is gauge-invariant modulo 2π , depends only on the path C , and is rate-independent.

Claim 3. The Berry curvature $\Omega_n = \nabla_{\mathbf{R}} \times \mathcal{A}_n$ is FORCED as the gauge-invariant curvature of the Berry connection, and the Stokes-theorem relation $\gamma_n[C] = \iint_S \Omega_n \cdot d\mathbf{S}$ holds for any surface S with $\partial S = C$ on which the bundle is trivializable.

Claim 4. The structural identity between Berry-phase machinery on parameter space and gauge-field machinery on physical space is FORCED by the rule-type connection being a single substrate primitive instantiated on different manifolds. Berry phase is the parameter-space holonomy of the same rule-type connection that produces gauge phase as physical-space holonomy.

Claim 5 (negative). No new substrate primitives are required. The Berry connection, Berry phase, and Berry curvature decompose into composition of: parameter-dependent participation rule (P-BP-1), identity alignment under parameter change (P-BP-2), adiabatic evolution as slow parameter variation (P-BP-3), the parameter-space bundle structure (P-BP-4), and closed loops in parameter space (P-BP-5). Plus standard differential-geometry machinery (gradient, exterior derivative, Stokes' theorem) re-derived as needed.

Claim 6 (scope-limit). This walkthrough does not derive: the non-Abelian (Wilczek–Zee) Berry connection for degenerate subspaces; sub-adiabatic corrections; the Aharonov–Anandan non-cyclic generalization; the Sjöqvist–Pati geometric phase for mixed states; or the Chern quantization of integrated Berry curvature on closed surfaces. Those items are flagged OPEN.

The unified statement. The Berry phase is the parameter-space holonomy of the rule-type connection on the bundle of parameter-dependent participation rules. The Berry connection, Berry curvature, and Berry phase are FORCED by adiabatic identity preservation acting on a smoothly parameter-dependent family of rules; they are not formal QM constructions but substrate-level rule-type geometry expressed in parameter space.

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